

Capstone Project: Enterprise Data Governance & Analytics Pipeline

AWS Glue & Athena Bench Training — Francheska Olympia

This document highlights the information processes used and followed to successfully complete this capstone project.

Dataset

[eCommerce behavior data from multi category store](#) — from Kaggle.

The months October and November 2019 were used to ingest data into the AWS S3 service as csv files.

For data evaluation, another raw data was ingested to properly test the Data Quality Definition Ruleset that will be used in the script.

testing-dataset.csv:

event_time	event_type	product_id	category_id	category_code	brand	price	user_id	user_session
2019-10-01 8:02:00	view	44600062	2103807459595387724	computers.notebook	lenovo	-500.25	535871217	72d76fde-8bb3-4e00-8c23-a032dfed738c
2019-11-05 9:02:00	view	17200506	2053013559792632471	computers.notebook	acer	-50000	541312140	7c90fc70-0e80-4590-96f3-13c02c18c713
2019-10-01 8:02:00	view	44600062	2103807459595387724	electronics.smartphone	apple	-68990	550050854	7c90fc70-0e80-4590-96f3-13c02c18c713
2019-11-01 8:02:00	view	17200506	2103807459595387724	electronics.smartphone	apple	75990		c6bd7419-2748-4c56-95b4-8cec9ff8b80d

- Negative `price` values were placed here to test the condition needed
- A `user_id` value was left null to test the condition needed

Data Schema

Data schema shows the metadata of the dataset. It serves as the blueprint to understand the data in the Medallion architecture. From raw (bronze) to conformed (silver) and up to curated (gold).

Database name: `bronze-cheska-capstone-db`

Table name: `01_bronze`

Column name	Data type	Comment
<code>event_time</code>	string	The timestamp of the user interaction
<code>event_type</code>	string	Categorized user interaction
<code>product_id</code>	bigint	Unique identifier for the product
<code>category_id</code>	bigint	Unique identifier for the category

category_code	string	The code of the category
brand	string	Brand name of the product associated with the category
price	double	Item price of the product
user_id	bigint	Unique identifier for the user
user_session	string	Session of the user associated with the event

Database name: silver-cheska-capstone-db

Table name: passed

Column name	Data type	Comment
event_time	timestamp	The timestamp of the user interaction
event_type	string	Categorized user interaction
product_id	bigint	Unique identifier for the product
category_id	bigint	Unique identifier for the category
category_code	string	The code of the category
brand	string	Brand name of the product associated with the category
price	double	Item price of the product
user_id	bigint	Unique identifier for the user
user_session	string	Session of the user associated with the event

Database name: silver-cheska-capstone-db

Table name: quarantined

Column name	Data type	Comment
event_time	timestamp	The timestamp of the user interaction
event_type	string	Categorized user interaction
product_id	bigint	Unique identifier for the product
category_id	bigint	Unique identifier for the category
category_code	string	The code of the category

brand	string	Brand name of the product associated with the category
price	double	Item price of the product
user_id	bigint	Unique identifier for the user
user_session	string	Session of the user associated with the event

Database name: gold-cheska-capstone-db

Table name: dim_product

Column name	Data type	Comment
product_id	bigint	Unique identifier for the product (Primary key)
category_id	bigint	Unique identifier for the category
category_code	string	The code of the category
brand	string	Brand name of the product associated with the category

Database name: gold-cheska-capstone-db

Table name: fact_events

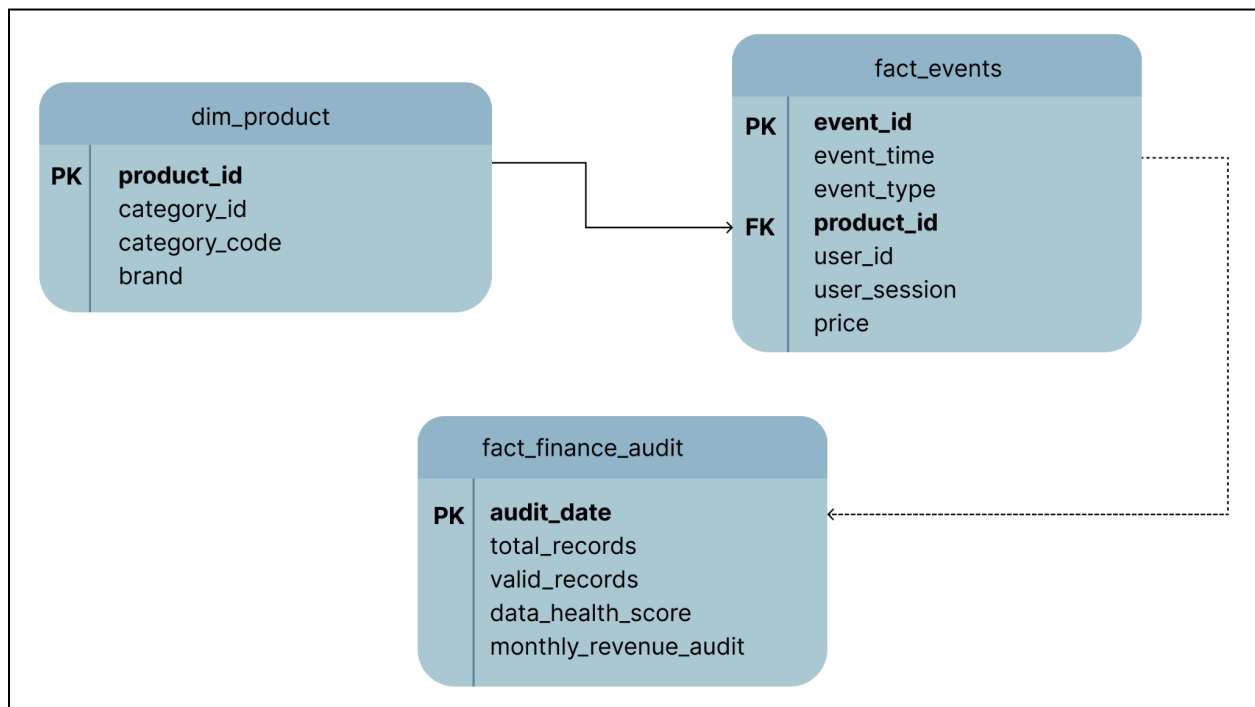
Column name	Data type	Comment
event_id	bigint	Event transaction identifier for the fact table
event_time	timestamp	The timestamp of the user interaction
event_type	string	Categorized user interaction
product_id	bigint	Foreign key reference to dim_product
user_id	bigint	Unique identifier for the user
user_session	string	Session of the user associated with the event
price	double	Item price of the product

Database name: gold-cheska-capstone-db

Table name: fact_finance_audit

Column name	Data type	Comment
audit_date	date	First day of the month representing the audit period
total_records	bigint	Total record count processed for the month
valid_records	bigint	Count of records containing valid price
data_health_score	double	Percentage score of valid records relative to total records
monthly_revenue_audit	decimal	Total monetary revenue sum computed across all valid events for the month

Data Model

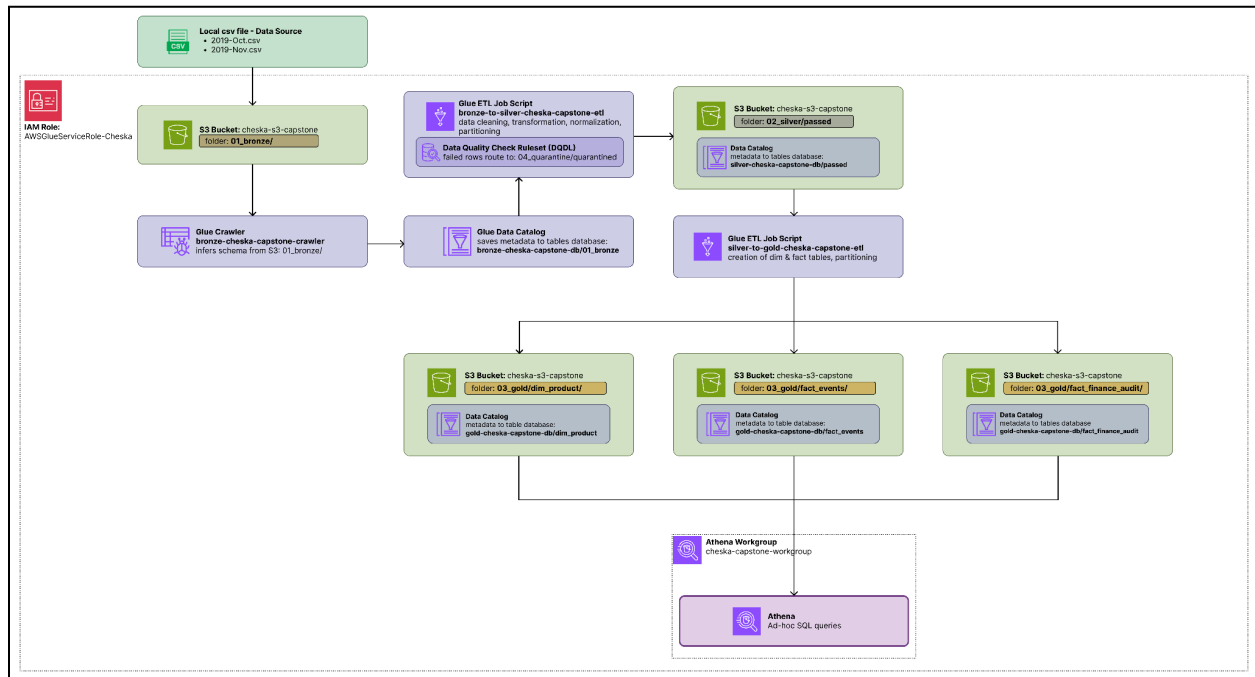


Upon creating the curated layer, Star schema model was established to properly organize the data according to the business needs and descriptions. The data model consists of one dimensional table and two fact tables.

- **dim_product** – provides clean, deduplicated product attributes for an organized, business reporting information
- **fact_events** – An atomic fact table that captures the user - interaction events

- **fact_finance_audit** – An aggregated fact table that was created from `event_time` of `fact_events`. It delivers the monthly financial revenue and data health metrics for financial business requirements

Architecture Diagram

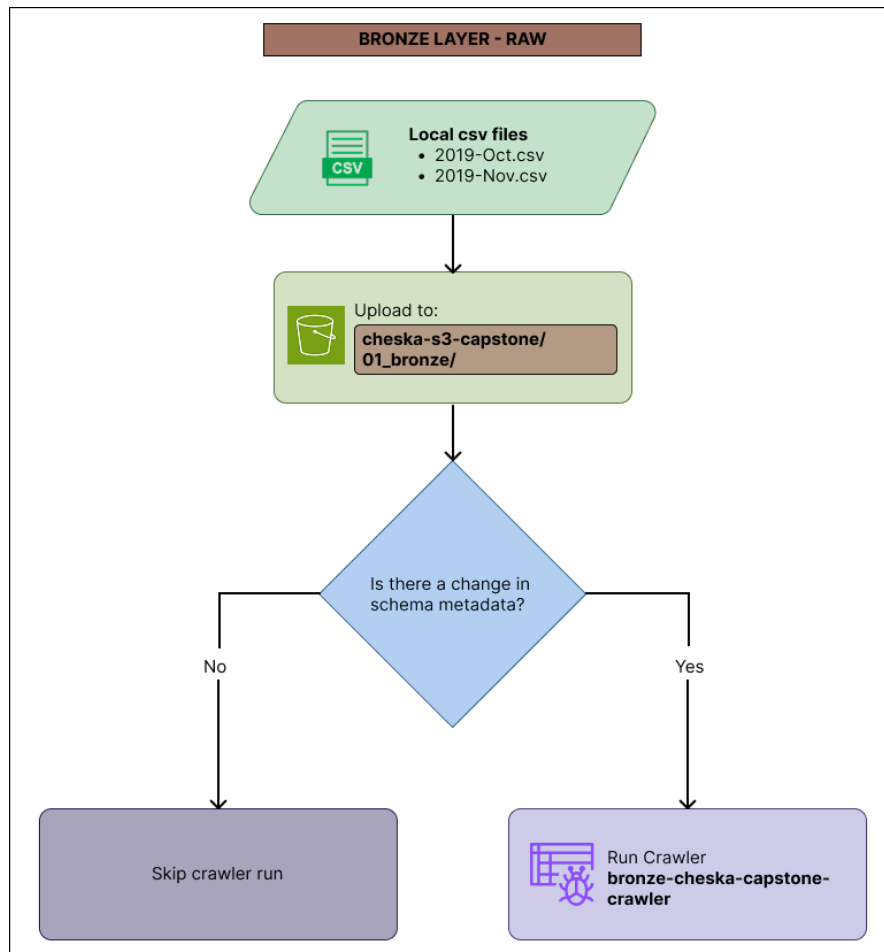


The architecture diagram illustrates how the whole data pipeline was done from data ingestion to raw layer up until the curated layer. Two local data csv files were ingested into the raw layer (bronze), once ingested, an IAM role `AWSGlueServiceRole-Cheska` was created to perform actions inside the whole data pipeline process in the environment. A crawler was then made to run and infer the schema from S3 bucket `cheska-s3-capstone/01_bronze/`, creating and placing the schema inside the database `bronze-cheska-capstone-db` in table `/01_bronze`. After the process of getting the metadata. A Job script `bronze-to-silver-cheska-capstone-etl` was started to perform data cleaning, transformation, normalization, and data evaluation was done in this process, saved to Iceberg format for automatic table creation in glue. Failed rows from data evaluation were routed to S3 bucket `cheska-s3-capstone/04_quarantine/quarantined`. For the passed rows, routed to `cheska-s3-capstone/02_silver/passed`, and database `silver-cheska-capstone-db`, table `/passed`. Job script `silver-to-gold-cheska-capstone-etl` was started to create the final stage of the data. Creation of dim and fact tables were created in this script all routed to different S3 buckets and database/tables. After successfully creating the curated (gold) layer, ad-hoc queries can be done in Athena in `cheska-capstone-workgroup`.

Design Decisions

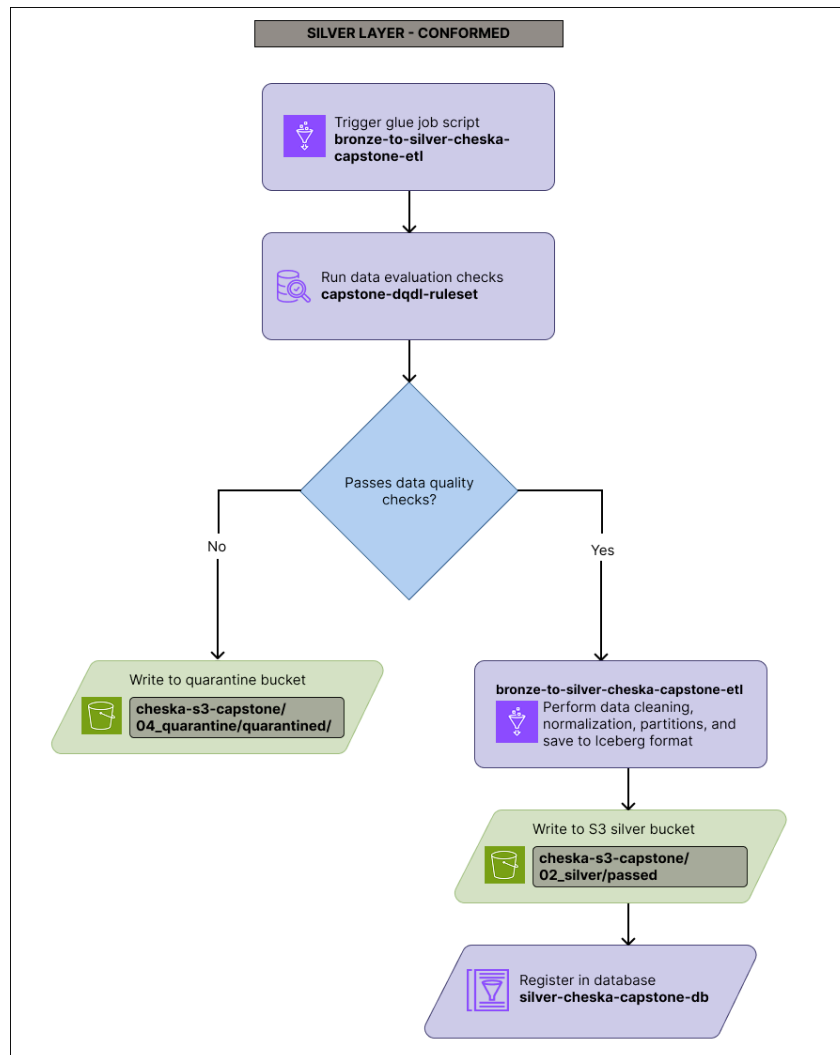
Decisions made with the whole pipeline flow were categorized into three sections according to the medallion architecture.

Bronze Stage Design Decision



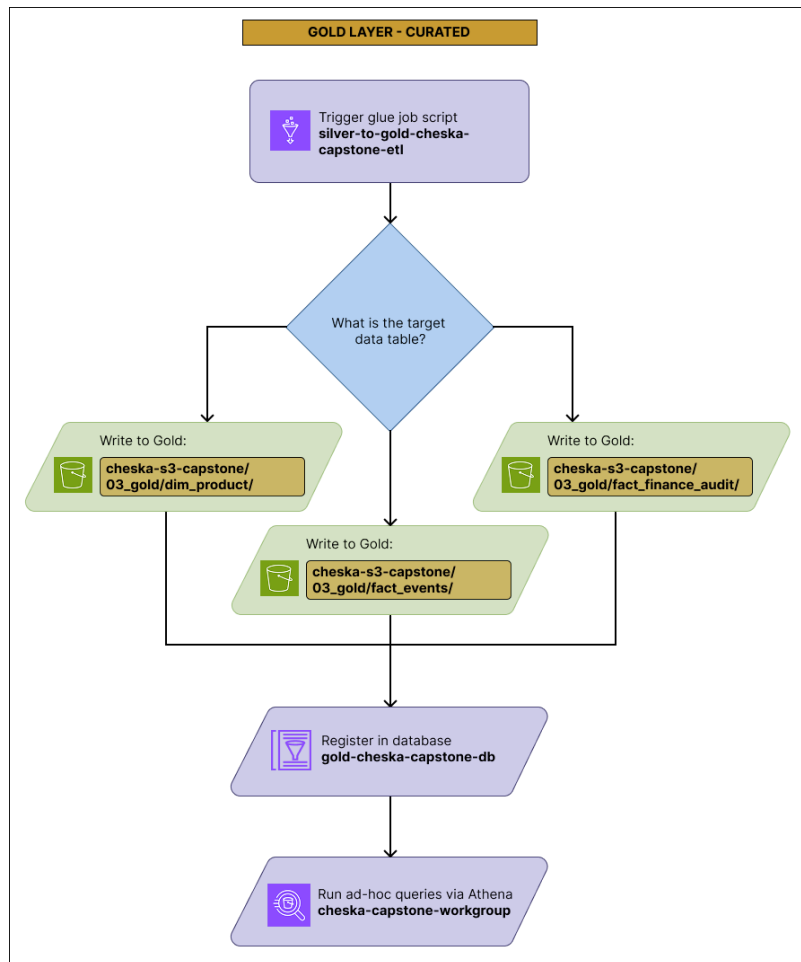
Data ingestion begins when the local csv data was uploaded to the S3 bucket bronze folder `cheska-s3-capstone/01_bronze`. The next step is to check if the schema of the data has changed, is new, or not. This is where the Glue crawler run `bronze-cheska-capstone-crawler` takes place, it infers the schema into the database `bronze-cheska-capstone-db` that will be used further on the next stage.

Silver Stage Design Decision



Continuing the bronze stage, glue job script **bronze-to-silver-capstone-etl** will be triggered to process the raw data and perform the whole data cleaning process. It first does data evaluation checks on the quality of the data using the ruleset **capstone-dqdl-ruleset** created and routes the passed and failed dataset into different branch flow. Negative data evaluation results will automatically proceed to the quarantine bucket **cheska-s3-capstone/04_quarantine/quarantined/**. While passed data will proceed to the next process which is data cleaning, normalization, partitioning the data and saving it into iceberg format for dynamic features, directly to S3 silver bucket **cheska-s3-capstone/02_silver/passed/** and automatically registering to database **silver-cheska-capstone-db**.

Gold Stage Design Decision



With iceberg data format, automatically registers or creates databases and tables in Glue, the gold stage starts with the last Glue job script process. The job script `silver-to-gold-cheska-capstone-etl` is triggered to perform the data model classification and routes to different branches. Dimension records fall into the S3 bucket `cheska-s3-capstone/03_old/dim_product/`. Fact tables of the events to `cheska-s3-capstone/03_old/fact_events/`, and lastly Fact tables for the finance audit data quality to `cheska-s3-capstone/03_old/fact_finance_audit/`. All S3 gold data automatically registers to glue databases and creates tables per data model because of the dynamic feature of Iceberg file format. Once populated, queries can be done using Athena under `cheska-capstone-workgroup` created.

The `fact_finance_audit` table was created as a dedicated aggregated table for access of finance analysts and data governance officers. It provides a summary of their needed information regarding the data that makes querying on their end simpler.

AWS Resources

Resource Name	AWS Service
cheska-s3-capstone	S3 bucket
bronze-cheska-capstone-db	Glue - Database
bronze-cheska-capstone-crawler	Glue - Crawler
capstone-dqdl-ruleset	Data Quality Definition Language Ruleset
bronze-to-silver-cheska-capstone-etl	Glue - Job
silver-cheska-capstone-db	Glue - Database
silver-to-gold-cheska-capstone-etl	Glue - Job
gold-cheska-capstone-db	Glue - Database
cheska-capstone-workgroup	Athena Workgroup
AWSGlueServiceRole-Cheska	IAM Role

Athena Queries

Checking of Quarantined Data in Quarantined table

Script:

SQL

```
SELECT * FROM "silver-cheska-capstone-db"."quarantined" LIMIT 5;
```

Result:

#	event_time	event_type	product_id	category_id	category_code	brand	price	user_id	user_session
1	2019-11-05 09:02:00.000000 UTC	view	17200506	2053013559792632471	computers.notebook	acer	-50000.0	541312140	7c90fc70-0e80-4590-96f3-13c02c18c713
2	2019-11-01 08:02:00.000000 UTC	view	17200506	2103807459595387724	electronics.smartphone	apple	75990.0		c6bd7419-2748-4c56-95b4-8cec9ff8b80d
3	2019-10-01 08:02:00.000000 UTC	view	44600062	2103807459595387724	computers.notebook	lenovo	-500.25	535871217	72d76fde-8bb3-4e00-8c23-a032dfed738c
4	2019-10-01 08:02:00.000000 UTC	view	44600062	2103807459595387724	electronics.smartphone	apple	-68990.0	550050854	7c90fc70-0e80-4590-96f3-13c02c18c713

Checking of Quarantined Data in Gold fact_events table

Script:

SQL

```
SELECT * FROM "gold-cheska-capstone-db"."fact_events" WHERE "price" < 0 LIMIT 5;
```

Result:

Results (0)

Copy

Download results CSV

Q Search rows

< 1 >

#	event_id	event_time	event_type	product_id	user_id	user_session	price
---	----------	------------	------------	------------	---------	--------------	-------

No results

Run a query to view results

Checking of Gold layer - dim_product

Script:

SQL

```
SELECT * FROM "gold-cheska-capstone-db"."dim_product" LIMIT 5;
```

Result:

Results (5)						
Search rows						
#	product_id	category_id	category_code	brand		
1	1002536	2053013555631882655	electronics.smartphone	apple		
2	1002923	2053013555631882655	electronics.smartphone	huawei		
3	1003105	2053013555631882655	electronics.smartphone	leagoo		
4	1003171	2053013555631882655	electronics.smartphone	sony		
5	1003333	2053013555631882655	electronics.smartphone	apple		

Checking of Gold layer - fact_events

Script:

```
SQL
SELECT * FROM "gold-cheska-capstone-db"."fact_events" LIMIT 5;
```

Result:

#	event_id	event_time	event_type	product_id	user_id	user_session	price
1	120259084288	2019-10-09 14:18:23.000000 UTC	view	1005160	516594224	b2be9a07-d360	230.89
2	120259084289	2019-10-09 14:18:23.000000 UTC	view	27800175	534968868	e335fc94-916e-	133.85
3	120259084290	2019-10-09 14:18:23.000000 UTC	view	28717989	558150614	0124c418-242b	122.27
4	120259084291	2019-10-09 14:18:23.000000 UTC	view	9800273	514530111	41c057ec-9813-	32.43
5	120259084292	2019-10-09 14:18:23.000000 UTC	view	1004873	519274280	dc8f7e6d-6222-	379.57

Checking of Gold layer - fact_finance_audit

Script:

```
SQL
SELECT * FROM "gold-cheska-capstone-db"."fact_finance_audit" LIMIT 5;
```

Result:

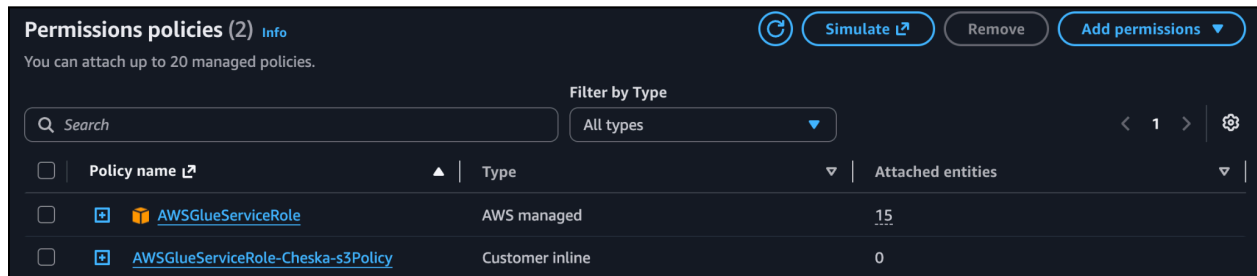
Results (2)							 Copy	 Download results CSV
 Search rows							< 1 > 	
#	audit_date	total_records	valid_records	data_health_score	monthly_revenue_audit			
1	2019-10-01	42448764	42448764	100.0	12323880556.04			
2	2019-11-01	67501979	67501979	100.0	19741582645.11			

IAM Role

Together with the capstone project and training, creation of an IAM role - policies is required. The role created provides access to buckets and AWS Glue actions.

IAM Role name: **AWSGlueServiceRole-Cheska-s3Policy**

Permission policies attached:



☐	Policy name ↗	Type	Attached entities
☐	 AWSGlueServiceRole	AWS managed	15
☐	 AWSGlueServiceRole-Cheska-s3Policy	Customer inline	0

AWSGlueServiceRole-Cheska-s3Policy:

```
JSON
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "Statement1",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:PutObject",
        "s3:DeleteObject"
      ],
      "Resource": [
        "arn:aws:s3:::cheska-s3-medallion/*", // the hands-on resource
        "arn:aws:s3:::cheska-s3-capstone/*"  // the capstone resource
      ],
      "Condition": {
        "StringEquals": {
          "aws:ResourceAccount": "[resource_account]"
        }
      }
    }
  ]
}
```